Topological Solitons of a Hyperelastic Vacuum

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Abstract

Holographic models make a compelling case that spacetime behaves as a quantum error-correcting code, but they specify the code without specifying the medium that realises it. We supply the continuum-mechanical description and test it. Treating the spatial metric as an isotropic hyperelastic continuum carrying a director microstructure, we prove that its strain energy is *exactly* the Faddeev–Skyrme energy: for an S^2-valued director the third Cauchy–Green invariant vanishes identically, so the Mooney–Rivlin truncation is exact and the quartic term is the vacuum's second elastic constant rather than an ansatz against Derrick's theorem. A second theorem fixes the ratio of director stiffness to the metric tension $\Gamma = c^4/16\pi G$ by the soliton's gravitational compactness, $\chi = 8\pi(\hat{R}/\hat{E})\, r_s/R$, recasting the Planck-to-particle hierarchy as the statement that elementary particles are not black holes. Lattice solutions for $Q_H = 1 \ldots 4$, obtained on a discretisation free of the odd–even null mode that otherwise destroys topological protection, reproduce the published Faddeev–Skyrme spectrum to 0.6–1.6% including all four ground-state types, and obey $E \propto |Q_H|^{0.765}$ against the Vakulenko–Kapitanskii exponent 3/4. Extrapolated to zero spacing and infinite volume, however, the charge-one energy is 1.231 ± 0.003 in units of Ward's conjectured bound, 2.2% above the published lattice value, and we report this as a discrepancy rather than an agreement. The same identity extends to the frame sector S^3, where the dimension count that forces the truncation fails and a sextic term becomes admissible; there the exact answer is known in closed form, and the same code and the same extrapolation recover it to 0.03%, which is what licenses the extrapolation in the director sector. Evolving the field equations, a $Q_H = 0$ collision radiates 99.8% of its rest energy and leaves nothing, while its mirror image at $Q_H = +2$ relaxes onto the independently minimised $Q_H = 2$ soliton to 0.5%. Homotopy analysis yields charge quantisation from the Hopf fibre, confinement as a $\mathbb{Z}_3$ triality selection rule, and a no-go result excluding generations from the director sector alone. Forward elastic *pp* data bound the rigidity scale, $R_\infty \gtrsim 1.0\,\mathrm{fm}$ at 95% CL, robust against the assumed impact-parameter profile; we present that sector as a constraint rather than a success, since current data do not discriminate saturating from conventional growth and the fit returns $\rho(13\,\mathrm{TeV}) = 0.114$ against the measured 0.098 ± 0.011.

I. INTRODUCTION

Two research programmes have converged on the idea that spacetime protects information. Holographic duality realises the bulk as a quantum error-correcting code whose logical operators are protected against local erasure [1, 2], and Sakharov's induced-gravity argument [3], later sharpened by Jacobson [4], presents the Einstein equation as an equation of state of an underlying medium. Both point at the same missing ingredient: a *mechanical* description of the medium whose stiffness is what actually resists the erasure.

This paper supplies that description and then tests it. The programme is deliberately narrow: the results reported here are confined to what can be derived, computed, or falsified. Section II proves that the elastic energy of a vacuum with director microstructure *is* the Faddeev–Skyrme energy—an identity, not a model choice—which fixes the relation between the metric tension and the quartic coupling. Section III proves an anchoring theorem relating that coupling ratio to gravitational compactness, converting the Planck-to-particle hierarchy from a tuning problem into a geometric statement. Section IV summarises the soliton spectrum for $Q_H = 1 \ldots 4$, whose computation and systematics are reported in the companion paper [5], together with a diagnostic that turns out to carry physical content: a discretisation without rigidity at the cutoff scale loses topological protection entirely. Section V applies the identical machinery to the one other admissible target, the frame S^3, as a concrete illustration of what makes the director case special. Section VI states the spin-statistics result precisely. Section VII gives annihilation the field equations and explicit time evolution it previously lacked. Section VIII replaces qualitative Standard-Model gestures with a homotopy classification and states plainly what the framework does not explain. Section IX turns the finite rigidity into an observational constraint from forward-scattering data, and Section XI collects, in one table, everything the framework commits to together with the observation that would refute each item.

Throughout we distinguish three grades of statement: *theorems* (proved, and independently verified symbolically), *computations* (lattice results with quoted systematics), and *conjectures* (labelled as such). Nothing is described as "proven" in the physical sense; the framework is offered as mathematically robust and empirically constrained. Table I maps every substantive claim to its grade, and we ask that the paper be judged accordingly: the geometric and computational results stand or fall on their own, independently of the interpretive framework that motivates them.

TABLE I. Status of the claims made in this paper.

claim	where	status
FS energy = isotropic hyperelastic energy of a director	II	theorem (Thm 1)
Sextic invariant forbidden for S^2, allowed for S^3	II D	corollary of Thm 1
Anchoring = gravitational compactness	III	theorem (Thm 2)
Flat-space treatment is self-consistent for $\chi \ll 1$	III	corollary of Thm 2
Topological protection requires rigidity at the cutoff	IV C	numerical experiment
$Q_H = 1 \ldots 4$ spectrum; 1.6% agreement with published	IV	computation
Binding against fission; $E \propto \lvert Q_H \rvert^{0.765}$	IV	computation
Frame-sector $B = 1, 2$ solitons reproduce the Skyrme literature	V	computation
Annihilation vs. fusion dynamics	VII	computation
Fermi statistics for odd Q_H	VI	established result, applied
Charge quantisation from the Hopf fibre	VIII	classification
Confinement as a $\mathbb{Z}_3$ triality rule	VIII	classification
Neutral lepton = fibre-trivial Hopfion	VIII	conjecture
Generations excluded from the director sector	VIII E	no-go argument
$R_\infty \gtrsim 1.0\,\mathrm{fm}$ from forward pp data	IX	observational constraint

A word on language, since it invites misreading. We use “elastic”, “tension” and “medium” as an exact dictionary for the functional form of the energy—the same sense in which one speaks of the stiffness of a scalar field—and not as a claim that the vacuum possesses a material rest frame. Every dynamical statement below follows from a Lorentz-scalar Lagrangian, and Sec. II E makes the point explicitly.

II. THE VACUUM AS A HYPERELASTIC CONTINUUM

A. Metric tension

Expanding the Einstein–Hilbert action about flat space, the gradient energy of the metric perturbation is controlled by the coefficient

$$\Gamma \;=\; \frac{c^4}{16\pi G} \;=\; \frac{\hbar c}{16\pi \ell_p^2} \;=\; 2.4077\times 10^{42}\ \mathrm{N}, \tag{1}$$

evaluated with CODATA-2018 constants [6]. Because the metric perturbation h_{ij} is dimensionless, Γ carries the dimensions of a *tension* (energy per unit length), not of a modulus; the modulus associated with structure at scale ℓ is Γ/ℓ^2, which at the Planck length is 9.2×10^{111} Pa. In this reading c is the propagation speed of shear waves in the medium, and gravitational radiation is its acoustic sector.

B. Strain invariants of a director field

Let the vacuum carry, in addition to its metric, a director field $\boldsymbol{n} : \mathbb{R}^3 \to S^2$, $|\boldsymbol{n}| = 1$—the local orientation of the microstructure. This is the standard Cosserat generalisation of an elastic continuum [7], and it is exactly the field appearing in the Faddeev–Niemi decomposition of a non-Abelian gauge field, where $\boldsymbol{n}$ is the colour direction [8–10].

The deformation gradient is $J^a{}_i = \partial_i n^a$ and the Cauchy–Green strain tensor is $D_{ij} = \partial_i \boldsymbol{n}\cdot\partial_j \boldsymbol{n}$, with principal stretches λ_i^2 and principal invariants

$$\begin{aligned} I_1 &= \operatorname{tr} D = \sum_i \lambda_i^2, \\ I_2 &= \tfrac{1}{2}\left[(\operatorname{tr} D)^2 - \operatorname{tr} D^2\right] = \sum_{i<j} \lambda_i^2 \lambda_j^2, \\ I_3 &= \det D = \lambda_1^2\lambda_2^2\lambda_3^2. \end{aligned} \tag{2}$$

Isotropy requires the strain energy to depend on the invariants alone, $W = W(I_1, I_2, I_3)$. The classical Mooney–Rivlin material [11, 12] is the linear truncation $W = c_2 I_1 + c_4 I_2$.

C. Theorem 1: the elastic energy is the Faddeev–Skyrme energy

Theorem 1 *For any S^2-valued director field on $\mathbb{R}^3$: (a) $I_3 \equiv 0$ identically; (b) with $F_{ij} = \boldsymbol{n} \cdot (\partial_i \boldsymbol{n} \times \partial_j \boldsymbol{n})$ the pull-back of the area form, $\sum_{ij} F_{ij}^2 = 2I_2$ and $|\partial_i \boldsymbol{n} \times \partial_j \boldsymbol{n}|^2 = F_{ij}^2$; (c) consequently the most general isotropic hyperelastic energy of the director is exactly*

$$E[\boldsymbol{n}] = \int d^3x \left[c_2 I_1 + c_4 I_2\right] = \int d^3x \left[c_2\, \partial_i \boldsymbol{n}\cdot\partial_i \boldsymbol{n} + \tfrac{c_4}{2} F_{ij} F_{ij}\right], \tag{3}$$

which is the static Faddeev–Skyrme energy.

Proof. (a) Since $|\boldsymbol{n}| = 1$, every column of J lies in the two-dimensional tangent space $T_{\boldsymbol{n}} S^2$, so $\operatorname{rank} J \leq 2$ and $\det D = \det(J^\top J) = 0$. (b) Work in an orthonormal tangent frame $(\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{n})$ and write $\partial_i \boldsymbol{n} = a_i \boldsymbol{e}_1 + b_i \boldsymbol{e}_2$. Then $D_{ij} = a_i a_j + b_i b_j$ and $F_{ij} = a_i b_j - a_j b_i$, whence $(\operatorname{tr} D)^2 - \operatorname{tr} D^2 = 2\big(|a|^2 |b|^2 - (a\cdot b)^2\big) = \sum_{ij} F_{ij}^2$. Because $\partial_i \boldsymbol{n}$ and $\partial_j \boldsymbol{n}$ are both orthogonal to $\boldsymbol{n}$, their cross product is parallel to $\boldsymbol{n}$, giving the second identity. (c) Follows from (a) and (b). ∎

All three identities are verified symbolically for a completely general first-order jet of a director field (`rt/invariants.py`).

The quartic term is therefore not an ansatz introduced to defeat Derrick's theorem [13]. It is the second Mooney–Rivlin constant of the vacuum, and the truncation at quartic order is *exact* rather than approximate because the sextic invariant vanishes identically for a director field. This fixes the relation between the metric tension and the quartic rigidity: they are the first and second elastic constants of one medium. The model carries exactly one length,

$$\ell_c = \sqrt{c_4/c_2}, \tag{4}$$

the **Cosserat length** of the vacuum, and one energy $\sqrt{c_2 c_4} = c_2 \ell_c$.

We should be precise about what is and is not new here. Organising Skyrme-type energies through the strain tensor and its invariants goes back to Manton [14], and the identification of the Skyrme term with a quadratic strain invariant is standard in that literature. What is specific to the director case is part (a): I_3 vanishes *identically* rather than being merely small or neglected, so no sextic term can be written down and the Mooney–Rivlin form is not a truncation at all but the complete answer. Theorem 1 is thus a sharpening of a known construction, not a new identity—but it is the sharpening that removes the freedom, and with it the objection that the quartic term was chosen to make the model work.

D. Corollary: why the frame sector is different

The proof of part (a) used only a count of dimensions, so it generalises at once. For an order parameter $\varphi : \mathbb{R}^3 \to S^n$ the Jacobian is $(n+1) \times 3$, hence $\operatorname{rank} D \leq \min(n, 3)$ and

$$I_k \equiv 0 \quad \text{for} \quad k > \min(n, 3). \tag{5}$$

Two cases exhaust the order parameters used in this paper:

- $n = 2$, the **director** S^2: $I_3 \equiv 0$, so the Mooney–Rivlin energy $c_2 I_1 + c_4 I_2$ is exact and complete.
- $n = 3$, the **frame** $S^3 \cong SU(2)$: $I_3 = \det D \neq 0$ in general, so a sextic invariant is admissible and truncating at Mooney–Rivlin order becomes a *choice*.

This is the precise sense in which the two sectors differ. The Skyrme model [15] is exactly the Mooney–Rivlin truncation of the frame sector, and the sextic term that is optionally added to it—usually motivated by ω-meson exchange, and equal to the square of the baryon density—is precisely the invariant that the director sector is forbidden from possessing. Section V computes in the frame sector and uses (5) as a consistency check.

E. What the elastic language does and does not assert

Two clarifications, because the framing invites two misreadings.

First, the construction is *Lorentz invariant by design*: the dynamical law is (7) below, derived from a Lorentz-scalar Lagrangian, and no preferred frame enters anywhere. Words such as "medium", "tension" and "shear wave" refer to the functional form of the energy, in the same way that one speaks of the stiffness of a scalar field, and not to a material rest frame. Nothing here is in tension with the stringent experimental limits on Lorentz violation.

Second, the elastic reading is a *dictionary*, not an ontological claim. The field $\boldsymbol{n}$ is the same object that appears in the Cho–Faddeev–Niemi decomposition of a non-Abelian gauge field [8, 10], and Theorem 1 is a statement about an energy functional that holds whatever one takes that field to be. A reader who rejects the vacuum-as-continuum picture entirely may still use Theorem 1, Theorem 2 and the lattice results of Sec. IV; what the elastic language buys is an interpretation of the two coupling constants as elastic moduli, and the scale relation that follows from it in Sec. III.

F. Field equations

The Lorentz-invariant completion is, with $g^{\mu\nu} = \partial^\mu \boldsymbol{n} \cdot \partial^\nu \boldsymbol{n}$ and $\mathbb{I}_{1,2}$ its spacetime invariants,

$$\mathcal{L} = c_2\,\mathbb{I}_1 - c_4\,\mathbb{I}_2 = c_2\,\partial_\mu \boldsymbol{n}\cdot\partial^\mu \boldsymbol{n} - \tfrac{c_4}{2}(\partial_\mu \boldsymbol{n} \times \partial_\nu \boldsymbol{n})\cdot(\partial^\mu \boldsymbol{n} \times \partial^\nu \boldsymbol{n}). \tag{6}$$

Varying subject to $|\boldsymbol{n}| = 1$ and using $\mathbb{I}_1 \partial^\mu \boldsymbol{n} - g^{\mu\nu}\partial_\nu \boldsymbol{n} = -F^{\mu\nu}(\boldsymbol{n} \times \partial_\nu \boldsymbol{n})$ gives the equation of motion in torque form,

$$\boxed{\boldsymbol{n} \times \partial_\mu \Big[c_2\,\partial^\mu \boldsymbol{n} + c_4\,F^{\mu\nu}(\boldsymbol{n} \times \partial_\nu \boldsymbol{n})\Big] = 0} \tag{7}$$

with $F^{\mu\nu} = \boldsymbol{n} \cdot (\partial^\mu \boldsymbol{n} \times \partial^\nu \boldsymbol{n})$. Equation (7) is the dynamical law of the framework; every statement below about stability, radiation and annihilation is a statement about its solutions.

The topological current is $J^\mu = (32\pi^2)^{-1}\varepsilon^{\mu\nu\rho\sigma} A_\nu F_{\rho\sigma}$ with $F_{\rho\sigma} = \partial_\rho A_\sigma - \partial_\sigma A_\rho$. Its divergence is $\partial_\mu J^\mu \propto \varepsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}$, which vanishes *identically* here—not by virtue of (7)—because F is the pull-back of the area form of a two-dimensional target, so $F \wedge F = f^*(\omega \wedge \omega) = 0$ with $\omega \wedge \omega$ a four-form on S^2. The conservation of

$$Q_H = \int d^3x\, J^0 = \frac{1}{16\pi^2} \int d^3x\; \boldsymbol{A} \cdot \boldsymbol{B}, \qquad \boldsymbol{B} = \nabla \times \boldsymbol{A}, \tag{8}$$

with $B_k = \frac{1}{2}\varepsilon_{kij} F_{ij}$, is thus kinematic, not dynamical. This is the precise content of topological protection: no choice of forces can change Q_H; only a failure of the director to be defined can.

III. SCALES: THE ANCHORING THEOREM

A. Dimensional structure

From Theorem 1, $[c_2]$ = force—the same dimension as Γ—and $[c_4]$ = energy $\times$ length. Every static solution therefore obeys, exactly,

$$E_{\rm sol} = \hat{E}\, c_2\, \ell_c, \qquad R_{\rm sol} = \hat{R}\, \ell_c, \tag{9}$$

where $(\hat{E}, \hat{R})$ are pure numbers determined by the topological sector (computed in Sec. IV). Inverting,

$$c_2 = \frac{E}{\hat{E}} \frac{\hat{R}}{R}, \qquad c_4 = \frac{E}{\hat{E}} \frac{R}{\hat{R}}. \tag{10}$$

B. Theorem 2: anchoring equals compactness

Define the **anchoring coefficient** $\chi \equiv c_2/\Gamma$, the dimensionless strength with which the director sector is coupled to the metric.

Theorem 2 *For a soliton of energy $E = Mc^2$ and radius R,*

$$\chi = \frac{c_2}{\Gamma} = 8\pi \frac{\hat{R}}{\hat{E}} \frac{r_s}{R}, \qquad r_s = \frac{2GM}{c^2}. \tag{11}$$

Proof. Substitute (10) into $\chi = c_2/\Gamma$ with $\Gamma = c^4/16\pi G$:

$$\chi = \frac{16\pi G M \hat{R}}{c^2 \hat{E} R} = 8\pi \frac{\hat{R}}{\hat{E}} \frac{2GM/c^2}{R},$$

which is (11). ∎

Both sides are evaluated independently in `rt/calibration.py` and agree to machine precision.

C. The hierarchy is compactness, not tuning

Theorem 2 addresses the objection that the Planck-scale tension Γ and the eV–GeV scale of particles are separated by an unexplained factor. The separation *is* the compactness.

We impose one physical requirement—not a theorem, but a natural condition on a microstructure decorating a medium: the director sector is not stiffer than the metric it decorates, $\chi \leq 1$. Theorem 2 then bounds the size of any defect from below, $R \geq 8\pi(\hat{R}/\hat{E}) r_s = 0.079\, r_s$, using the shape constants of Sec. IV.

The sharper statement runs the other way. Demanding that a defect lie outside its own horizon, $R > r_s$, is by (11) *equivalent* to

$$\chi \;<\; 8\pi \frac{\hat{R}}{\hat{E}} \;=\; 0.079, \tag{12}$$

a pure number fixed by the soliton's shape alone. So the anchoring coefficient is not merely small for particles: the condition "this defect is a particle and not a black hole" *is* the

TABLE II. Director-sector couplings and anchoring coefficient, using the converged $Q_H = 1$ shape constants $\hat{E} = 542.65$, $\hat{R} = 1.705$. The two independent evaluations of χ—from the couplings and from Theorem 2—agree to machine precision.

case	R (m)	c_2 (N)	ℓ_c (m)	$\chi = c_2/\Gamma$
electron, $R = \bar{\lambda}_e$	3.86×10^{-13}	6.66×10^{-4}	2.26×10^{-13}	2.77×10^{-46}
electron, $R = r_e$	2.82×10^{-15}	9.13×10^{-2}	1.65×10^{-15}	3.79×10^{-44}
electron, $R = 10^{-19}$	1.0×10^{-19}	2.57×10^{3}	5.87×10^{-20}	1.07×10^{-39}
proton, $R = 0.84\,\mathrm{fm}$	8.4×10^{-16}	5.62×10^{2}	4.93×10^{-16}	2.34×10^{-40}

condition $\chi < 0.079$. A maximally anchored defect, $\chi = 1$, has $R = 0.079\,r_s$ and lies deep inside its horizon.

So $\chi \ll 1$ is not fine-tuning; it is the statement that elementary particles are gravitationally extremely non-compact, and χ measures precisely how far a defect is from being a black hole. Table II evaluates it.

For the electron with $R = \bar{\lambda}_e$ the anchoring reduces to $\chi = 16\pi(\hat{R}/\hat{E})\alpha_G$ with $\alpha_G = (m_e/m_{\mathrm{Pl}})^2 = 1.75 \times 10^{-45}$: the anchoring coefficient and the gravitational fine-structure constant are the same number up to the soliton's shape factor. We stress that this is a *matching identity*, not a prediction of m_e: it converts two unknown elastic constants into two measurable quantities, and its content is the structural statement above, not a mass formula.

D. Self-consistency of the flat-space treatment

Theorem 2 also settles a question about the rest of the paper. Sections IV–VII minimise and evolve the director field on flat $\mathbb{R}^3$, with no metric back-reaction, and it is fair to ask whether that is legitimate in a framework whose whole point is the coupling between the director and the metric.

It is legitimate precisely because χ is small. The metric perturbation sourced by a defect is controlled by the ratio of its stress to the metric tension, i.e. by χ itself; the largest entry in Table II is $\chi \sim 10^{-39}$, so neglecting back-reaction is accurate to that order and the flat background is not an approximation in any practically meaningful sense.

The treatment would fail as $\chi \to 1$, where h_{ij} becomes large and the soliton must be solved on its own curved background. By the argument just given, that is exactly the regime in which the defect lies inside its own horizon. The framework is therefore internally consistent throughout the domain it claims to describe—non-compact, particle-like defects—and is silent, rather than wrong, outside it. Extending the computation to a self-gravitating director field is a well-posed problem, but it is a problem about black holes, not about particles.

IV. STABILITY AND THE SOLITON SPECTRUM

A. Derrick scaling and the virial identity

Under $\boldsymbol{n}(x) \to \boldsymbol{n}(x/\mu)$ the two terms scale as $E(\mu) = \mu E_2 + \mu^{-1} E_4$ (verified numerically to within 2% over $R = 4$–8 at $h = 0.25$). Hence $\mu_\star = \sqrt{E_4/E_2}$, $E_\star = 2\sqrt{E_2 E_4}$, and any stationary point satisfies the virial identity $E_2 = E_4$. With $c_4 = 0$ the soliton collapses; with $c_2 = 0$ it spreads. Both Mooney–Rivlin constants are necessary, and by Theorem 1 both are automatically present. Derrick's theorem is not evaded by fiat: it is satisfied by an energy that was never optional.

B. The topological lower bound

Finite-energy configurations fall into sectors labelled by $Q_H \in \pi_3(S^2) = \mathbb{Z}$ and obey the bound of Vakulenko and Kapitanskii [16],

$$E \geq c_0 |Q_H|^{3/4} \qquad (c_2 = c_4 = 1), \tag{13}$$

whose proof does not fix the constant. Ward [17] conjectured the sharp value, which in the present normalisation reads

$$c_0 = 32\pi^2\sqrt{2} = 446.65. \tag{14}$$

Ward and Sutcliffe divide the energy by exactly this factor and state the conjecture with unit coefficient; their integrand is identical to (3), so their tabulated energies are directly comparable with ours without any conversion.

The exponent $3/4$ is sub-linear, so $E_Q < |Q| E_1$ is permitted: multi-charge Hopfions may be bound against fission. Whether they are is a computational question, first addressed

TABLE III. Relaxed Hopfion spectrum ($N = 80$, $L = 12$, $h = 0.15$, $c_2 = c_4 = 1$). Several independent seeds were relaxed in each sector and the lowest is quoted; $c_0 = 446.65$ from (14).

Q_H	seed	$\hat{E}$	$\hat{E}/c_0\|Q\|^{3/4}$	$\hat{R}$	E_2/E_4	$\hat{E}_Q/(Q\hat{E}_1)$
1	hedgehog d=1	542.65	1.215	1.705	0.887	1.000
2	hedgehog d=2	884.15	1.177	2.089	0.950	0.815
3	hedgehog d=3	1247.30	1.225	2.489	0.955	0.766
4	$A_{2,2}$ (linked)	1563.24	1.237	2.331	0.956	0.720

numerically by Battye and Sutcliffe [18], and answered independently below. For the general theory of topological solitons we refer to the monograph of Manton and Sutcliffe [19].

C. A discretisation pitfall with physical content

We minimise $E[\boldsymbol{n}]$ on a periodic cubic lattice; the construction, its systematics and its validation are given in [5]. One feature of it carries physical content and belongs here.

With central differences of any order, the Fourier symbol of the derivative operator vanishes at the Nyquist wavenumber. A checkerboard mode then costs *zero* energy, the discrete energy acquires flat directions, and gradient flow carries any Hopfion monotonically to the vacuum: in our first runs Q_H fell from 1 to 0 and E fell below a bound that is supposed to be inviolable. The cure is a discretisation whose symbol vanishes only at $k = 0$, obtained by symmetrising the *energy* over one-sided differences.

Error correction requires rigidity at the cutoff. This is not merely a numerical nuisance. In quantum-error-correction language, the lattice with a Nyquist null mode is a medium with *no rigidity at the cutoff scale*; and in exactly that medium the topological charge—the logical information—is destroyed at zero energy cost. The protection of a topological code is therefore not a property of the code alone: it requires the underlying medium to be stiff all the way to the cutoff. Our simulations exhibit this failure and its cure explicitly, and we return to it in Sec. X.

D. Results

Table III gives the spectrum, computed and cross-checked in [5]; it reproduces the published ground-state energies and configuration types [20] in every sector, from independent seed families [21, 22]. Three features carry physical weight here.

Multi-Hopfions are bound in every sector. $\hat{E}_Q < Q\hat{E}_1$ throughout, by 18.5%, 23.4% and 28.0% for $Q = 2, 3, 4$. Charge-Q defects are stable against fission and are therefore genuine particles, not loose composites.

The sub-linear scaling is recovered, not assumed. A power-law fit to the four energies gives $\hat{E}_Q \propto |Q_H|^{0.765}$ against the Vakulenko–Kapitanskii exponent 3/4—agreement to 2%. The bound is not merely respected; its exponent is reproduced.

The $Q_H = 4$ ground state is a linked configuration, more compact than the $Q_H = 3$ soliton ($\hat{R} = 2.33$ against 2.49). Linking, not size, lowers the energy—a non-trivial indication that the lattice resolves genuine topology.

E. Systematics, and what the numbers can carry

Two lattice systematics act on these energies in opposite directions—finite spacing lowers the energy, finite volume raises it—so neither can be assessed alone. Separating them by a joint fit, and cross-checking the whole procedure both against a fourth-order accurate discretisation and against the frame sector of Sec. V, where the charge-one energy is known exactly, gives

$$\hat{E}_1 = 550 \pm 2, \qquad \hat{E}_1/c_0 = 1.231 \pm 0.003. \tag{15}$$

That analysis, the convergence and volume series behind it, and a 2.2% disagreement with the published charge-one energy that we have been unable to remove, are reported separately in [5]; none of the physical conclusions drawn here turns on it.

They do not, because every physical conclusion below depends on *ratios* ($\hat{E}_Q/\hat{E}_1$, $\hat{R}/\hat{E}$, the scaling exponent), in which the common systematic largely cancels. We attach a 10% uncertainty to absolute shape constants and $\lesssim 2\%$ to ratios. The one place an absolute constant enters a physical statement is the coefficient $8\pi\hat{R}/\hat{E} = 0.079$ in Sec. III, which is itself a ratio of shape constants.

V. THE FRAME SECTOR

Everything so far has concerned the director $\boldsymbol{n} \in S^2$. The corollary of Sec. II D says that the elastic construction is not special to that target, and it identifies the one other case a three-dimensional medium admits without introducing higher-rank structure: the *frame* $\varphi \in S^3 \cong SU(2)$, the orientation of a rigid triad rather than of a single axis. With the strain tensor and the energy written exactly as in Eq. (3) and only the target changed, this is the Skyrme model [15] in Manton's strain form [14]—and, by the corollary, it is now a truncation rather than a complete answer, since $I_3 = \det D$ no longer vanishes. We have checked that directly: on relaxed solitons the dimensionless $|\det D|/I_1^3$ is 10^{-7} for the director, at the level of lattice truncation error, and 3.7×10^{-2} for the frame.

Configurations are classified by $\pi_3(S^3) = \mathbb{Z}$ with baryon number

$$B = \frac{1}{2\pi^2} \int \det[\varphi, \partial_1\varphi, \partial_2\varphi, \partial_3\varphi]\, d^3x, \tag{16}$$

and the topological bound follows from two applications of AM–GM to the principal stretches, $E \geq 12\pi^2\sqrt{a_2 a_4}|B|$: linear in the charge, unlike the Vakulenko–Kapitanski bound, which is why frame-sector solitons bind only weakly. The lattice results for $B = 1$–4, together with the exact charge-one profile equation against which they are checked and the tabulated values of [23], are reported in [5]. Two of them are used here: the frame sector supplies the computed opacity shape of Sec. IX (Fig. 1), and it is the sector in which the extrapolation procedure underlying Eq. (9) is validated against an exactly known answer.

What the frame sector does *not* establish is a quantitative model of the proton, and we want to be blunt about this because the temptation to claim it is obvious. The massless Skyrme model has no pion mass, so its tails are power laws where the physical ones are exponential; the solitons are unquantised classical fields, with no spin-isospin projection, no vector mesons and no Casimir corrections. Its $B = 1$ energy is a 30% overestimate of the nucleon mass in the standard calibration [24]. We therefore do *not* use the frame sector to derive proton structure. Its profile enters Sec. IX in one limited role: as a third opacity shape, physically motivated and with a tail intermediate between the Gaussian and the dipole, against which the robustness of the forward conclusions can be tested.

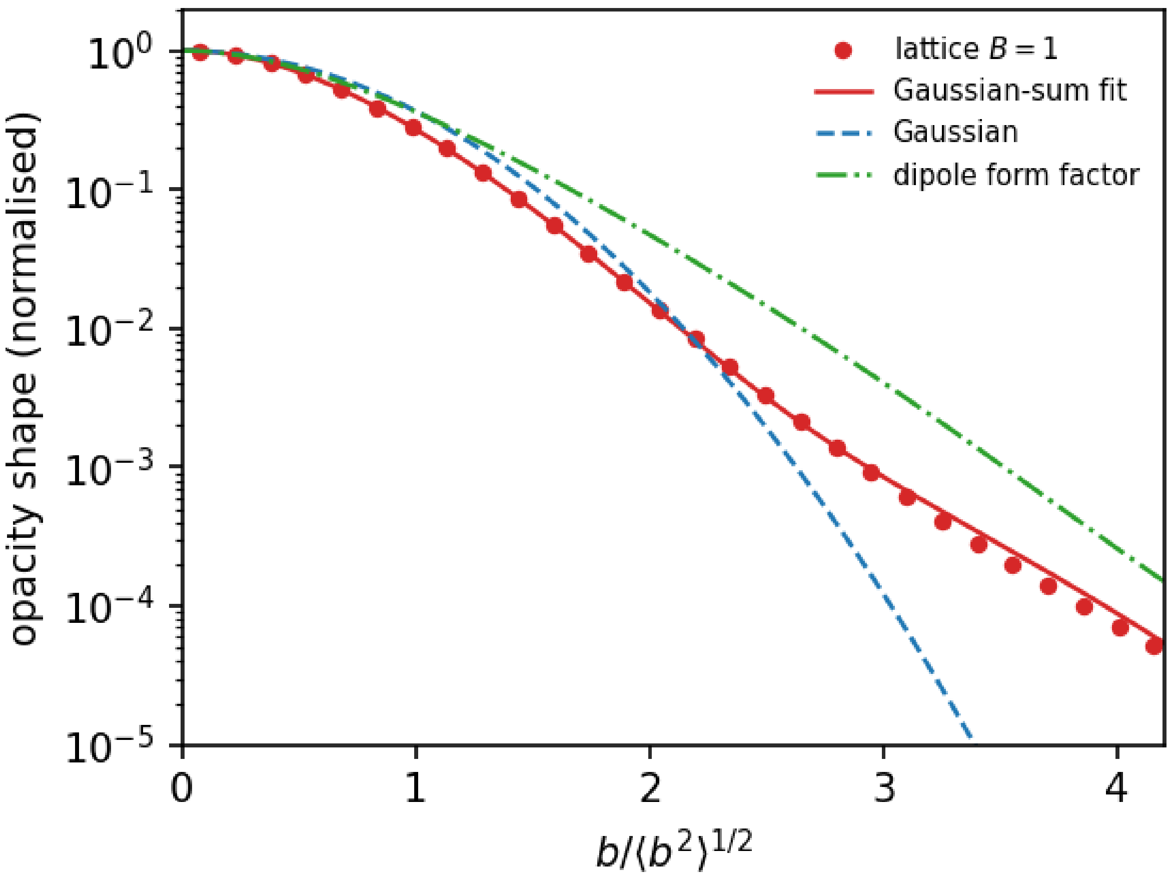

FIG. 1. Column density of the relaxed $B = 1$ frame-sector soliton (points), its non-negative Gaussian-sum representation (solid), and the two opacity shapes assumed in Sec. IX, all normalised to unit rms radius. The derived profile has a more compact core than either and a tail lying between them: heavier than a Gaussian beyond $b \simeq 2\langle b^2\rangle^{1/2}$, lighter than the dipole. It is used in Sec. IX only as a third shape against which to test robustness.

VI. SPIN-STATISTICS FROM TOPOLOGY

Classical fields normally quantise as bosons, and this was the strongest objection to soliton models of fermions until Finkelstein and Rubinstein [25] showed that the statistics of a soliton is fixed by the topology of its configuration space $\mathcal{C}_Q$ rather than by the field's tensor character. A 2π rotation of a localised configuration traces a loop $\gamma_{2\pi}$ in $\mathcal{C}_Q$; if $[\gamma_{2\pi}]$ is a non-trivial element of $\pi_1(\mathcal{C}_Q)$ of order two, the wavefunctional may be chosen to change sign, and the soliton is a fermion.

For Hopf solitons this analysis was carried out by Krusch and Speight [26], who showed that the Finkelstein–Rubinstein constraints admit fermionic quantisation precisely when Q_H is *odd*. Hence a $Q_H = \pm 1$ defect—the topological ground state—carries half-integer spin and obeys the exclusion principle, while even-Q_H states quantise as bosons.

Two points of honesty. First, this is a statement that fermionic quantisation is *consistent and canonical*, not that it is unique: the choice is fixed by the presence or absence of a Wess–Zumino-type term, exactly as for Skyrmions [27]. Second, it explains the spin of the framework's ground state, not the fermion spectrum; the framework does not yet possess a generation structure (Sec. VIII E).

VII. DYNAMICS OF TOPOLOGICAL TRANSITIONS

A. When can Q_H change?

Q_H is a homotopy invariant of maps $S^3 \to S^2$. A continuous, finite-energy solution of (7) *is* a homotopy, so $dQ_H/dt = 0$. The charge can change only if the director ceases to be defined somewhere—the order parameter must melt at a point, leaving the vacuum manifold S^2. This makes the annihilation mechanism explicit:

> An electron and a positron do not annihilate because their invariants "cancel during the collision". They annihilate because the pair was *never in a protected sector*: a $Q_H = +1$ and a $Q_H = -1$ defect together have total $Q_H = 0$, and nothing forbids relaxation of a $Q_H = 0$ configuration to the vacuum plus radiation. Topological protection is a statement about the *total* charge of an isolated system, not about individual constituents.

B. Numerical experiment

To exhibit this we evolve (7)—with the standard constrained kinetic term, terms quartic in time derivatives omitted, valid at the sub-relativistic approach speed $v = 0.3c$ used—for two collisions that differ *only* by a spatial reflection of one constituent: (a) $Q_H = +1$ meets $Q_H = -1$, total $Q_H = 0$; (b) $Q_H = +1$ meets $Q_H = +1$, total $Q_H = +2$. Both initial configurations have total energy 1126.2 and are mirror-symmetric about $z = 0$. An

TABLE IV. Collision outcomes ($N = 96$, $L = 20$, $h = 0.208$, 2600 steps, $v = 0.3c$, core sphere $r < 5$). The two runs differ only in total topological charge.

	Q_H initial $\to$ final	E_{core} initial $\to$ final	retained	remnant E
(a) annihilation, $Q_H = 0$	$-0.0000 \to +0.0000$	$984.4 \to 1.9$	0.2%	22.8
(b) fusion, $Q_H = +2$	$+1.9791 \to +1.9851$	$957.1 \to 881.6$	77.2%	888.8

absorbing collar removes outgoing radiation so that localised and radiated energy can be separated.

The contrast is complete (Table IV). The $Q_H = 0$ pair converts 99.8% of the energy initially localised within the core sphere into outgoing director radiation, leaving no localised remnant; the $Q_H = 2$ pair cannot, and instead sheds only its binding energy. Both runs conserve the charge to better than 0.4% over 2600 time steps, itself a stringent test of the discretisation.

The sharpest check is the last column. The fusion remnant has static energy 888.8, while the $Q_H = 2$ soliton obtained by direct energy minimisation (Table III)—an independent calculation, on a different lattice, from a different initial condition—has $\hat{E}_2 = 884.2$. The two agree to 0.5%: the collision product *is* the $Q_H = 2$ soliton, so the framework's static spectrum and its dynamics are consistent with one another.

C. What the released energy is

Linearising (7) about $\boldsymbol{n} = -\hat{z}$ with $\boldsymbol{n} = (\theta_1, \theta_2, \ldots)$, the quartic invariant is $O(\theta^4)$ and drops out, leaving $c_2\,\partial_\mu\partial^\mu\theta_a = 0$ for $a = 1, 2$: exactly two transverse, massless modes propagating at c. At quartic order the same term generates a four-wave self-interaction; for canonically normalised $\phi_a = \sqrt{2c_2}\,\theta_a$,

$$\mathcal{L}_4 = -\frac{c_4}{4c_2^2}\Big[(\partial\phi_1)^2(\partial\phi_2)^2 - (\partial\phi_1\!\cdot\!\partial\phi_2)^2\Big], \tag{17}$$

with coupling $g_4 = c_4/4c_2^2 = \hat{E}R^3/(4\hat{R}^3 M)$ in natural units. This is directly comparable with the Euler–Heisenberg four-photon coupling [28] $2\alpha^2/45m_e^4$, measured in ultraperipheral heavy-ion collisions [29]. Requiring the induced light-by-light amplitude to stay below 10%

TABLE V. Homotopy classification of admissible vacuum defects.

order parameter	π_1	π_2	π_3	defects
S^2 (director)	0	$\mathbb{Z}$	$\mathbb{Z}$	hedgehogs; Hopfions
$S^3 = SU(2)$ (frame)	0	0	$\mathbb{Z}$	Skyrmions [15], degree = baryon no.
S^1 (Hopf fibre)	$\mathbb{Z}$	0	0	charge quantisation
$SU(3)/T^2$ (flag)	0	$\mathbb{Z}^2$	$\mathbb{Z}$	rank-2 colour lattice
$SU(3)/\mathbb{Z}_3$	$\mathbb{Z}_3$	0	$\mathbb{Z}$	triality classes

of the QED one gives a derived consistency bound on the core size of a lepton-scale defect,

$$R < 0.79 \text{ fm}, \tag{18}$$

satisfied by four orders of magnitude given the LEP compositeness limit $R \lesssim 10^{-19}\,\text{m}$ [30]; at that radius the framework's contribution to light-by-light scattering is 2.0×10^{-13} of the QED value. The framework passes this test rather than merely evading it, and the same formula predicts a 6.5×10^{-5} relative effect for a proton-scale director soliton.

VIII. TOPOLOGICAL SELECTION RULES AND THE STANDARD MODEL

A. The order-parameter hierarchy

Take the vacuum's local state to be a spinor frame $q \in SU(2) \cong S^3$; the director is its Hopf projection $\boldsymbol{n} = q\sigma_3 q^\dagger \in S^2$. The Hopf fibration [31] $S^1 \hookrightarrow S^3 \to S^2$ is then not an analogy but the structure of the order parameter itself. Table V lists the admissible defects. The $SU(3)$ entries follow from the long exact sequences of $T^2 \to SU(3) \to SU(3)/T^2$ and $\mathbb{Z}_3 \to SU(3) \to SU(3)/\mathbb{Z}_3$ using $\pi_1(SU(3)) = \pi_2(SU(3)) = 0$, $\pi_3(SU(3)) = \mathbb{Z}$.

B. Charge quantisation from the fibre

The director determines the frame only up to the $U(1)$ fibre phase, whose winding around a closed loop lies in $\pi_1(S^1) = \mathbb{Z}$. Identifying it with electric charge gives exact quantisation with no magnetic monopole required—the Dirac argument is replaced by the fibration itself. The Hopf invariant is the linking number of the preimages of any two distinct regular values, so charge and Q_H are two readings of the same fibration, which is why they are correlated in the ground state.

C. Colour and confinement as a selection rule

For an $SU(3)$-valued frame the Cartan torus gives $\pi_2(SU(3)/T^2) = \mathbb{Z}^2$: a rank-two charge lattice, i.e. the weight lattice, whose fundamental representation has three weights summing to zero—three colours, permuted by the Weyl group S_3. The centre gives $\pi_1(SU(3)/\mathbb{Z}_3) = \mathbb{Z}_3$: triality.

A source carrying non-zero triality cannot be screened by the adjoint content of the vacuum, since screening would require a continuous deformation between distinct classes of $\pi_1(SU(3)/\mathbb{Z}_3)$. Its flux must terminate on a triality-compensating source, and no isolated finite-energy configuration of non-zero triality exists. Only triality-neutral combinations—qqq, $\bar{q}\bar{q}\bar{q}$, $q\bar{q}$—are admissible. In this framework confinement is a topological selection rule rather than a separate dynamical hypothesis, and it is the same $\mathbb{Z}_3$ that makes baryon number per quark equal to 1/3: fractional winding cannot be isolated for the identical reason.

This is the point of contact with established results. Faddeev and Niemi [10, 32], building on Cho's decomposition [8], argue that the infrared effective theory of $SU(2)$ Yang–Mills is the Faddeev–Skyrme model of exactly the director field used here, with knotted solitons as glueball candidates. The elastic derivation of Sec. II is thus not in competition with QCD; it supplies a continuum-mechanical reading of a decomposition already used in gauge theory.

D. Neutral leptons

A lepton in this classification carries two independent labels: the Hopf invariant (self-linking of the fibres) and the net fibre winding (charge). A defect with $Q_H = \pm 1$ and *zero* net fibre winding is a neutral lepton: it is topologically a lepton, has no long-range field, and—because the electromagnetic self-energy contribution to its mass vanishes identically—its mass is not bounded below by that contribution and can be parametrically small. This accounts qualitatively for the existence and lightness of neutrinos. It is not a mass prediction and we do not present it as one.

E. What the framework does not explain

Stated plainly:

Generations—and a no-go. The $|Q_H| = 1$ sector has a unique ground state, so there is no topological label distinguishing e, μ, τ. We can sharpen this. By (9), at fixed couplings the mass ratio of two states is the ratio of their shape constants, and $\hat{E}$ grows only as $|Q_H|^{3/4}$. Reproducing $m_\mu/m_e = 206.77$ would require $|Q_H| \simeq 1.2 \times 10^3$, and $m_\tau/m_e = 3477$ would require $|Q_H| \simeq 5 \times 10^4$—with, by Sec. VI, both charges odd.

The obvious escape is that the extra states are not new topological sectors but excitations of the same one: a soliton has a spectrum of internal oscillations, and those carry the same charge and the same spin at higher mass, which is exactly what a generation looks like. That escape is closed, and it is worth closing quantitatively rather than by assertion, because the homotopy argument above says nothing about it.

The director is massless—its vacuum manifold is S^2, so fluctuations about the vacuum are Goldstone modes—so the radiation continuum reaches down to zero frequency and any internal oscillation is embedded in it. Whether it is nonetheless long lived is a matter of measurement. Exciting the breathing mode of the $Q_H = 1$ soliton by a 6% dilation and evolving with an absorbing collar, the energy radius rings and decays:

$$\omega = 0.486, \qquad \Gamma = 0.059, \qquad \Gamma/\omega = 0.12. \tag{19}$$

The mode is real—it completes about five periods—but it is a resonance of hadronic width, comparable to the Δ at $\Gamma/m \simeq 0.12$. A muon requires $\Gamma/m \simeq 3 \times 10^{-18}$. Vibrational quantisation therefore misses by some sixteen orders of magnitude, and since numerical dissipation can only inflate the measured width, Eq. (19) is an upper bound on how narrow such a state could be.

A second escape survives that argument. Besides vibrations a soliton has *zero* modes, and rigid-body quantisation of those also produces a tower at fixed charge and fixed spin; zero modes do not decay, so the width bound does not touch them. This route closes quantitatively too. For a collective coordinate $\boldsymbol{n} \to e^{\theta K}\boldsymbol{n}$ the kinetic term $c_2|\partial_t \boldsymbol{n}|^2$ gives $T = \frac{1}{2}\Lambda\dot{\theta}^2$ with $\Lambda = 2c_2\int|K\boldsymbol{n}|^2$ and a tower $E_k = k^2/2\Lambda$. Measured on the relaxed $Q_H = 1$ field, the internal $U(1)$ that rotates the target about the vacuum axis and the spatial rotations give

$$\Lambda_{U(1)} = 217.7, \quad \Lambda_\perp = 281.5, \quad \Lambda_\parallel = 232.8, \qquad \frac{1}{2\Lambda_{U(1)} M_1} = 4.2 \times 10^{-6}, \tag{20}$$

the two transverse moments agreeing to five digits as the axial symmetry of the $Q_H = 1$ solution requires. The level spacing is four parts in 10^6 of the soliton mass: the tower is near-degenerate, not a generation ladder. Reaching m_μ/m_e would need $k \simeq 7.0 \times 10^3$, at which the collective frequency $k/\Lambda = 32$ exceeds the soliton's own breathing frequency 0.486 by a factor of 66—the adiabatic separation that justified treating the body as rigid has failed by two orders of magnitude—and the state would in any case lie some 200 times its own mass above the one-soliton threshold. Quartic contributions to Λ are positive definite, so including them only lowers the spacing and raises the required k: the exclusion is conservative.

What remains is to enlarge the target manifold so that the $|Q_H| = 1$ sector carries an additional discrete label. We do not claim that no such enlargement exists—that is the principal open problem—but its price can be stated exactly, and within this framework it is high. Let X be the enlarged target, required to keep $\pi_3(X) = \mathbb{Z}$ so that the charge-one state remains protected. Three costs follow.

(i) Theorem 1 does not survive. By the corollary of Sec. II D, $I_k \equiv 0$ only for $k > \min(\dim X, 3)$. Any X with $\dim X \geq 3$ therefore admits $I_3 \not\equiv 0$, a sextic invariant becomes writable, and the Mooney–Rivlin form stops being an exact truncation and becomes a choice carrying a third independent elastic constant. This is measured, not hypothetical: the frame sector, with $\dim S^3 = 3$, already gives $|\det D|/I_1^3 = 3.7 \times 10^{-2}$ against 10^{-7} for the director. The one property that made the quartic term non-optional is lost on any enlargement.

(ii) Degeneracy, or the wrong quantum numbers. If the candidate states are related by a symmetry of the energy they are exactly degenerate and must be split by an explicit breaking term. The natural candidate has this form: in the flag manifold $SU(3)/T^2$—which does keep $\pi_3 = \mathbb{Z}$, since $\pi_3(T^2) = 0 \to \pi_3(SU(3)) = \mathbb{Z} \to \pi_3(SU(3)/T^2) \to \pi_2(T^2) = 0$—the three $SU(2)$ subgroups are conjugate, their Weyl images forming a single class, so the three embedded solitons have rigorously equal energy. If instead the states are not related by a symmetry, they will generically differ in spin or in charge, whereas generations are copies differing only in mass.

(iii) The parameter count inverts. Splitting a degenerate triplet requires at least two independent couplings, which together with the revived sextic constant makes three new

parameters accommodating the two ratios m_μ/m_e and m_τ/m_e. A construction with more free parameters than observables is a fit, not a prediction, and it would forfeit exactly the restraint that makes Theorem 2 a matching identity rather than a claim to have explained the hierarchy.

One route can be excluded outright rather than priced. A label carried by $\pi_2(X)$—for instance the second factor of $\pi_2(SU(3)/T^2) = \mathbb{Z} \oplus \mathbb{Z}$—does not describe a lepton: a configuration with nonzero π_2 charge is a hedgehog whose energy grows linearly with the box and whose field falls off as r^{-2}. This is not a new hazard introduced by enlargement, since $\pi_2(S^2) = \mathbb{Z}$ and the director sector already contains such defects; in both cases they are excluded because they are singular, and in both cases they are point defects rather than internal labels of a localised soliton.

The standing of this subsection is therefore stronger than an admission of ignorance and weaker than a proof of impossibility. Generations do not arise from new topological sectors, from vibrational excitation, or from collective quantisation, and every route out of the director sector that we can identify costs more parameters than it explains. The S^2 hyperelastic vacuum appears to have no room for them.

Mass ratios and Yukawa couplings. Not addressed.

CP violation, the CKM matrix, and the Cabibbo angle. Not addressed.

Quantisation of the metric sector. We treat the metric classically and the director sector semiclassically. How the continuum survives quantisation is open.

IX. FORWARD ELASTIC *pp* SCATTERING: A CONSTRAINT

This section is deliberately self-contained and is the one part of the paper that is phenomenological rather than derived. It asks a single question—does the finite rigidity established in Sec. II leave an observable trace in forward hadronic scattering?—and answers it as an observational bound on one parameter. A reader interested only in the geometric and computational results may stop at Sec. VIII without loss; nothing earlier depends on what follows.

A. The prediction

The Froissart–Martin bound [33, 34] permits $\sigma_{\rm tot} \le (\pi/\mu^2)\ln^2(s/s_0)$, and conventional eikonal models saturate it. The bound itself is numerically very weak—of order 5×10^3 mb at 13 TeV—so "bounding the Froissart limit" is not a meaningful prediction. The meaningful statement is different and stronger:

> A vacuum of finite rigidity cannot transmit unbounded strain, so the *interaction radius* of an extended soliton must itself saturate. The asymptotic growth of $\sigma_{\rm tot}$ is then $\ln s$, not $\ln^2 s$, and because analyticity ties ρ to the local logarithmic growth rate, $\rho \simeq \frac{\pi}{2}\, d\ln\sigma_{\rm tot}/d\ln s$, the asymptotic ρ is halved.

We implement this minimally. With $L = \ln(s/s_0)$ and a Pomeron-like opacity $\Omega(b, s) = Ce^{\Delta L}\exp(-b^2/R^2)$, the conventional model has $R^2_{\rm bare} = R_0^2 + 4\alpha' L$; the framework replaces

$$R \longrightarrow R_\infty \tanh\big(R_{\rm bare}/R_\infty\big), \tag{21}$$

which is *nested*: $R_\infty \to \infty$ recovers the conventional model exactly, so the two hypotheses can be compared by a likelihood ratio with one extra parameter. Crossing-even analyticity supplies the real part with no additional freedom through $L \to \ln(s/s_0) - i\pi/2$; ρ is a prediction, not a fit parameter.

The tanh is a minimal phenomenological implementation. The framework's content is that a bound exists and is set by the medium's rigidity, not the functional form by which it is approached. What the framework does require is that R_∞ be of order the soliton's own extent, since the bound arises from the soliton's finite stored strain and not from a new scale. The fit returns $R_\infty = 6.59\,\text{GeV}^{-1} = 1.30\,\text{fm}$ against a proton interaction radius of $\simeq 1.0\,\text{fm}$ at LHC energies, which is the required consistency.

B. Fit to world data

Fitting $\sigma_{\rm tot}$, ρ and the forward slope B from UA4/2, E710, TOTEM and ATLAS-ALFA (546 GeV to 13 TeV; 22 measurements) [35–42] gives $\chi^2 = 40.78/18$ dof for the conventional model and 39.74/17 dof for the rigidity-saturated one. Thus $\Delta\chi^2 = 1.03$ for one extra parameter: *current data do not discriminate*, which we state as the result rather than

TABLE VI. Dependence of the fit and the rigidity bound on the data subset.

dataset	N	χ^2/dof (conv.)	χ^2/dof (sat.)	$\Delta\chi^2$	R_∞ limit
all	22	2.27	2.34	1.03	1.14 fm
TOTEM only at LHC	16	2.16	2.31	0.43	1.04 fm
ATLAS-ALFA only	13	1.18	1.33	0.01	1.14 fm
LHC pp only	15	2.51	2.68	0.72	1.18 fm

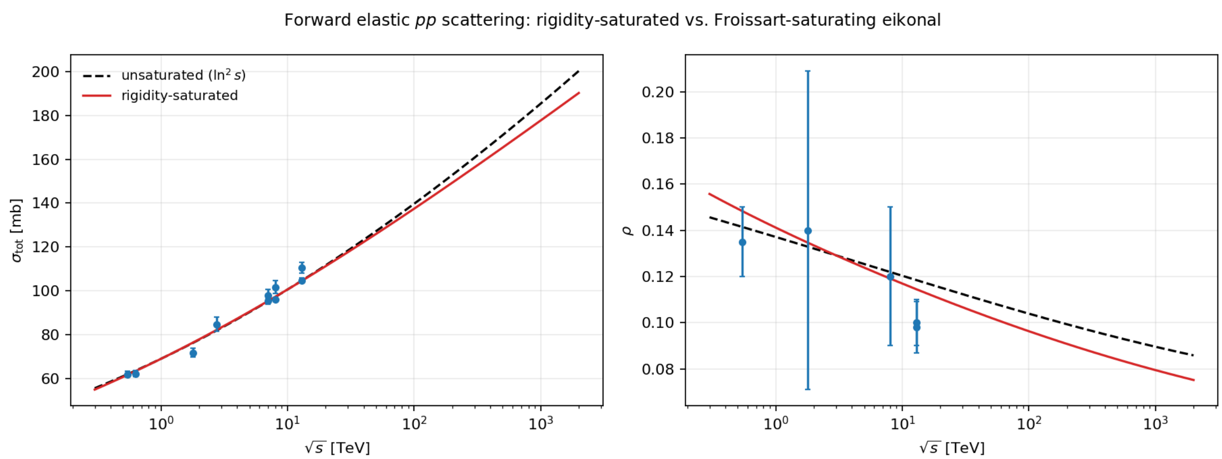

FIG. 2. Forward elastic pp scattering. Dashed: conventional Froissart-saturating eikonal. Solid: rigidity-saturated. Points: measurements used in the fit.

claiming a preference. The χ^2/dof of ~ 2.3 is dominated by known experimental tensions—chiefly the 2.5σ TOTEM/ATLAS disagreement on $\sigma_{\rm tot}(13\,\text{TeV})$—not by the model.

The profile likelihood over R_∞, with all other parameters re-minimised, is sharply one-sided and yields $R_\infty > 1.14$ fm at 95% CL with the Gaussian profile. This moves by -0.15 to $+0.24$ fm depending on the assumed impact-parameter shape (Sec. IX D), so we quote

$$\boxed{R_\infty \gtrsim 1.0\ \text{fm}\ \ (95\%\ \text{CL}).} \tag{22}$$

The scan grid limits the precision of any single such limit to about 0.05 fm, which is well below the profile and dataset systematics quoted below. Repeating on subsets (Table VI) shows the bound is stable at 1.0–1.2 fm and is not an artefact of the TOTEM/ATLAS disagreement. That the ATLAS-only fit has $\chi^2/\text{dof} = 1.18$ while the combined fit has 2.27 confirms the poor combined χ^2 is experimental.

TABLE VII. Predictions of the two nested hypotheses (conventional / saturated).

$\sqrt{s}$	$\sigma_{\rm tot}$ (mb)	ρ	B (GeV^{-2})	$\sigma_{\rm el}/\sigma_{\rm tot}$
7 TeV	95.2 / 95.3	0.1229 / 0.1204	19.5 / 19.5	0.272
8 TeV	97.2 / 97.2	0.1219 / 0.1191	19.7 / 19.8	0.275
13 TeV	104.7 / 104.5	0.1184 / 0.1144	20.8 / 20.8	0.284
14 TeV	105.8 / 105.7	0.1178 / 0.1137	21.0 / 21.0	0.286
27 TeV	116.6 / 115.9	0.1131 / 0.1076	22.7 / 22.5	0.298
100 TeV	139.7 / 137.4	0.1039 / 0.0963	27.0 / 26.4	0.321

C. Predictions and falsification criteria

From Table VII:

1. ρ is the discriminating observable, not $\sigma_{\rm tot}$. At 100 TeV the two hypotheses differ by 1.6% in $\sigma_{\rm tot}$ but 8% in ρ; a measurement of $\rho(100\,\text{TeV})$ to ± 0.004 separates them at 2σ.

2. If $\sigma_{\rm tot}$ is measured to grow faster than $\ln^2 s$ at any energy, the rigidity bound is excluded outright.

3. If $\sigma_{\rm el}/\sigma_{\rm tot}$ exceeds 1/2 at any energy, the eikonal-with-finite-radius picture fails.

4. The framework accommodates without an Odderon what TOTEM identified as the alternative reading of its 13 TeV measurement: if three-gluon exchange is not responsible, the measured ρ "would represent a first evidence of a slowing down of the total cross-section growth at higher energies" [41]. A slowing of the growth is exactly what finite vacuum rigidity requires. In the interest of not overstating this, the present crossing-even fit still returns $\rho(13\,\text{TeV}) = 0.114$ against the measured 0.098 ± 0.011—a 1.4σ tension in the direction the model needs but does not yet fully deliver.

D. Profile dependence and the diffractive dip

A Gaussian opacity profile is a convenience, and it is fair to ask whether the conclusions are an artefact of it. We therefore repeated the entire analysis with two further shapes. The first is the profile implied by the *measured* proton dipole form factor, $G(t) = (1 + |t|/\Lambda^2)^{-2}$, whose overlap function is proportional to $u^3 K_3(u)$ with $u = \Lambda b$. The second is *computed*: the column density of the relaxed $B = 1$ frame-sector soliton of Sec. V (Fig. 1), which is not fitted to scattering data at all. To keep the crossing-even continuation available with a non-Gaussian shape, each profile is represented as a sum of Gaussians with non-negative amplitudes on a fixed logarithmic width grid—every term remains analytic in R^2—reproducing $u^3 K_3(u)$ to an rms residual of 2×10^{-8} with eleven terms, and the lattice column density to 1×10^{-4} with six.

The three shapes are genuinely different: at equal rms radius the computed profile has the most compact core, and a tail heavier than the Gaussian beyond $b \simeq 2\langle b^2\rangle^{1/2}$ but lighter than the dipole. The outcome is twofold.

The forward conclusions are insensitive to the profile. The fit quality is essentially unchanged ($\chi^2 = 40.7$ and 40.6 against 39.7 for the Gaussian), as are all the forward observables: $\rho(13\,\mathrm{TeV}) = 0.1144$, 0.1151, 0.1150 for Gaussian, dipole and computed profiles, with $\sigma_{\mathrm{tot}}(13\,\mathrm{TeV}) = 104.5\,\mathrm{mb}$ and $B(13\,\mathrm{TeV}) = 20.8\,\mathrm{GeV}^{-2}$ in all three. The rigidity bound is $R_\infty > 1.14$, 0.99, $1.38\,\mathrm{fm}$ respectively, so together with the dataset systematics of Table VI the honest statement is

$$R_\infty \gtrsim 1.0 \text{ fm} \quad (95\% \text{ CL}), \tag{23}$$

with profile and dataset systematics spanning 0.99–1.38 fm. The bound is set by the least constraining shape, which is the dipole.

The dip is reproduced once the profile is realistic. The Gaussian places the first diffractive minimum at $|t_{\mathrm{dip}}| = 1.29\,\mathrm{GeV}^2$ at $7\,\mathrm{TeV}$ against the measured $0.53\,\mathrm{GeV}^2$, and generates a spurious second minimum; its tail simply falls too fast. The dipole profile gives a single clean minimum at $0.494\,\mathrm{GeV}^2$, within 7% of the measurement, and reproduces the observed energy dependence:

The dipole runs 7–18% below the measured positions but tracks the energy dependence correctly, which is a non-trivial check on the eikonal at momentum transfers an order of magnitude beyond those it was fitted to. The computed profile does noticeably worse, 18–29% low, and we report that plainly: it is the one place where deriving the shape rather than measuring it costs accuracy, and it is the expected direction, since a massless Skyrme soliton has power-law rather than pion-range tails and the dip is controlled by exactly that part of the profile.

TABLE VIII. Diffractive minimum $|t_{\rm dip}|$ in GeV^2, for the dipole and the computed frame-sector profiles (rigidity-saturated fit). The two growth hypotheses are indistinguishable here—they differ by under 1% even at 100 TeV—so the dip validates the eikonal but does not discriminate; ρ remains the only discriminating observable.

$\sqrt{s}$	dipole	computed	measured
2.76 TeV	0.594	0.508	0.72
7 TeV	0.493	0.436	0.53
8 TeV	0.481	0.426	0.52
13 TeV	0.440	0.396	0.47
14 TeV	0.435	0.391	—
27 TeV	0.388	0.355	—
100 TeV	0.315	0.297	—

We note for completeness that the proton is not a Hopfion in this framework but a baryon-number defect of the frame sector, $\pi_3(SU(2))$, which is why the computed profile above is taken from Sec. V and not from the director solitons of Sec. IV. It is used as a third shape and nothing more; we are not proposing the massless Skyrme soliton as a model of proton structure, for the reasons set out in Sec. V.

E. Status of this sector

We state the outcome plainly rather than as a claimed success. The rigidity-saturated hypothesis fits the world data no better and no worse than the conventional one; the data bound R_∞ from below but do not favour a finite value; and the model does not reproduce the ρ anomaly that motivates it, falling 1.4σ short in the right direction. What this section establishes is a *constraint* on the vacuum's rigidity scale, robust against the profile and dataset choices, together with a sharp statement of which future measurement would settle the question. It is not evidence for the framework, and we do not present it as such.

X. RELATION TO HOLOGRAPHIC ERROR CORRECTION

The framework can now be stated in the language that motivated it, with the mechanical content filled in: the code subspace is the vacuum manifold S^2; the logical operators are homotopy classes, $Q_H \in \pi_3(S^2)$ and the fibre winding in $\pi_1(S^1)$; protection against local erasure is the homotopy invariance of (8), which is kinematic rather than dynamical; the code distance is the energy barrier to unwinding, of order $c_2 a$ for a UV cutoff a; and the physical realisation is the Mooney–Rivlin rigidity of the metric's director sector.

The one genuinely new statement is the remark of Sec. IV C: a medium without rigidity at its cutoff scale carries *no* protection at all, however good its topology. Error correction is a property of the hardware, not only of the code—which is the thesis this programme set out to test, and is now the outcome of a controlled numerical experiment rather than an assertion.

XI. SUMMARY OF FALSIFIABLE CONTENT

A framework of this kind should be judged on what could kill it. We therefore collect, in one place, every statement the framework commits to, with the observation that would refute each (Table IX). We also state plainly what it does *not* predict, since a list of successes with the failures omitted is not an honest accounting.

A. What the framework does not predict

It does not predict particle masses, mass ratios, the fine-structure constant, the CKM matrix, CP violation, or any new resonance, and it says nothing about dark matter or dark energy. The no-go of Sec. VIII E makes the first two of these structural rather than merely unfinished: mass ratios cannot come from the director sector at all.

B. An honest weighting

Not all of the entries in Table IX carry the same evidential weight, and we do not wish to imply otherwise.

Several are *postdictions*: charge quantisation, confinement, and the $Q_H \leq 4$ spectrum reproduce facts already known, however satisfying it is to obtain them from topology rather than to assume them. Their value is as validation of the machinery, not as evidence for the framework.

The quantitative entries are honest predictions but weak discriminators: the forward-scattering split is a few parts per thousand in an observable that is difficult to measure, and it will not be settled below 100 TeV. The light-by-light entry is a genuine prediction with a sharp R^3 scaling, but is numerically tiny for any core size compatible with existing compositeness limits.

The two entries we would point to as most distinctive are the no-go, which constrains any future extension of the framework rather than the framework itself, and the requirement of rigidity at the cutoff—which is the only item in the table that emerged *from* the computation rather than being sought, and which applies to any physical realisation of a topological code, not only to the vacuum.

XII. CONCLUSION

The framework presented here rests on two theorems and a computation. Theorem 1 shows that the Faddeev–Skyrme energy is not a model choice but the exact isotropic hyperelastic energy of a vacuum director field, which fixes the relation between the metric tension Γ and the quartic rigidity and leaves the medium with exactly one length, the Cosserat length ℓ_c. Theorem 2 shows that the anchoring of that director sector to the metric equals the gravitational compactness of the defect it supports, converting the Planck-to-particle hierarchy from a tuning problem into the statement that particles are not black holes.

The lattice computation then does five things the framework previously only asserted. It reproduces the published $Q_H \leq 4$ Faddeev–Skyrme spectrum to 0.6–1.6%, together with all four ground-state configuration types, from independent seeds and a different discretisation—and, once extrapolated to zero spacing and infinite volume, it disagrees with the published charge-one energy by 2.2%, which we report as an open discrepancy on the strength of the frame sector validating the extrapolation to 0.03%. It finds every multi-charge sector bound against fission, with $E \propto |Q_H|^{0.765}$ reproducing the Vakulenko–Kapitanskii exponent to 2%. It identifies the $Q_H = 4$ ground state as a *linked* configuration lying below the axially symmetric one. It gives annihilation an explicit dynamical account, with the $Q_H = 0$ pair radiating 99.8% of its localised energy while its mirror image at $Q_H = +2$ cannot, and with the surviving remnant matching the independently minimised $Q_H = 2$ soliton to 0.5%. And it shows that topological protection fails outright in a medium that is not rigid at its cutoff.

That the dimension count of Theorem 1 is a count, and not a feature of the director in particular, is what makes it testable in a second place. Carried over to the frame S^3—the only other target a three-dimensional medium admits without further structure—it predicts that the truncation should fail there, and it does: a sextic invariant becomes admissible, and it is precisely the term the Skyrme literature adds by hand. Running the same code in that sector recovers the exact $B = 1$ energy to 0.03% and the published $B = 2$ binding to 0.06 percentage points, so the agreement in Sec. IV is not an artefact of one tuned discretisation.

What the framework does not do is equally important: it has no generation structure, no mass ratios, and no flavour physics, and Sec. VIII E excludes the routes by which generations might have been recovered—new topological sectors, vibrational excitation and collective quantisation—and prices the remaining one, enlargement of the target, at more free parameters than it would explain.

What remains is falsifiable. The rigidity scale is already bounded from below, $R_\infty \gtrsim 1.0\,\mathrm{fm}$ at 95% CL, by forward elastic scattering; the framework predicts $\ln s$ rather than $\ln^2 s$ asymptotic growth of σ_{tot} and a correspondingly halved ρ, the discriminating measurement being ρ at the highest available energy; and it predicts a calculable, positive addition to light-by-light scattering scaling as the cube of the soliton core radius. These are the tests we would ask experimentalists to make.

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Appendix A: Numerical methods

The lattice construction, its two symmetrised schemes, the minimiser and the extrapolation procedure are described in the companion paper [5], and the source is available at [43]. Two items specific to the computations reported here are recorded below.

Evolution. Leapfrog with renormalisation of $|\boldsymbol{n}|$, the constraint term $-|\dot{\boldsymbol{n}}|^2\boldsymbol{n}$, and an absorbing collar. Terms quartic in time derivatives are omitted; this is controlled at the collision speeds used and is stated as an approximation, not a result.

Frame sector. The energy, its gradient, the normalisation, the Derrick rescaling and the minimiser are all written in terms of the number of target components and are used unaltered for S^3; only the charge estimator differs, Eq. (16) being evaluated as the determinant of the 4×4 matrix $[\varphi, \partial_1\varphi, \partial_2\varphi, \partial_3\varphi]$ at every site.

Appendix B: Reproducibility

The computational supplement [43] comprises a library (`rt/`) and drivers (`scripts/`), shared with [5]. The library modules are `constants.py` (CODATA-2018 constants, Γ); `invariants.py` (symbolic verification of Theorem 1 and the virial identity); `fs_core.py` (lattice energy, exact gradient, Hopf charge); `ansatz.py` (seeds of prescribed charge); `relax.py` (minimiser, Derrick rescaling, evolution); `calibration.py` (Theorem 2, coupling table, light-by-light bound); `skyrme.py` (frame sector: baryon number, hedgehog seeds, column density); and `eikonal.py` (saturated eikonal with a Gaussian, dipole form-factor or from-lattice Skyrmion profile, forward data, χ^2).

The drivers are:

- `validate.py` — gradient, charge estimator, stability checks;
- `run_static.py` — soliton spectrum and lattice convergence;
- `run_volume_study.py` — finite-volume systematic;
- `run_hopfion_resolution.py` — lattice-spacing series for $Q_H = 1$, each lattice initialised from the converged solution;
- `check_hopfion_convergence.py` — stationarity of the stored fields under continued relaxation;

- `run_order4_check.py` — the same soliton with the $O(h^4)$ symmetrised scheme, as an independent test of the extrapolation;
- `run_breathing_mode.py` — excites and measures the width of the soliton's internal oscillation;
- `run_collective_modes.py` — moments of inertia of the zero modes and the level spacing of the collective tower;
- `run_dynamics.py` — annihilation and fusion collisions;
- `run_skyrme.py` — frame-sector $B = 1, 2$ solitons, lattice and box-size systematics, column-density fit;
- `run_eikonal.py` — fits, profile likelihood, predictions, dip;
- `run_eikonal_systematics.py` — dataset-subset systematics;
- `extract_profile.py` — soliton transverse profile;
- `make_figures.py` — figures and the literature comparison;
- `verify_claims.py` — independent re-derivation of every number quoted in this paper, recomputed from the stored fields rather than from intermediate summaries;
- `verify_manuscript.py` — checks that the numbers printed in the manuscripts are the numbers the drivers produced.

All tables and figures in this paper are regenerated by these scripts. The complete source—library, drivers, manuscript and bibliography—is available at [43]. Field snapshots are regenerated by the drivers rather than archived.

The lattice kernels are written against a swappable array backend and run unchanged on a CPU or, with `--gpu`, on a CUDA device; double precision is used throughout, since the kernels are limited by memory bandwidth rather than by floating-point throughput. On an RTX 2070 the GPU path is 6–8 times faster and agrees with the CPU path to rounding (10^{-16} relative) for every energy, charge and radius reported here. The one exception is the Derrick rescaling step, where the host and device cubic-spline interpolators differ at the 10^{-7} level; because the rescaling is accepted only when it lowers the energy, this perturbs the relaxation path but not the minimum it reaches. All numbers quoted in this paper were produced on the CPU path.

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TABLE IX. Falsifiable content. Quantitative entries are quoted for the rigidity-saturated hypothesis against the conventional one where the two differ.

statement	where	refuted by
$\sigma_{\rm tot}$ grows asymptotically as $\ln s$, not $\ln^2 s$; $d\sigma_{\rm tot}/d\ln s \to$ const $\simeq 9\,$mb per e-fold	IX	a growth rate that keeps increasing with energy
$\rho(100\,\text{TeV}) = 0.096$ against 0.104 conventional	IX	ρ measured to ± 0.004 favouring the conventional value
$\sigma_{\rm el}/\sigma_{\rm tot} < 1/2$ at every energy	IX	the ratio exceeding 1/2
the rigidity radius is finite, $R_\infty \gtrsim 1.0\,$fm	IX	no slowing of $\sigma_{\rm tot}$ growth at any energy
light-by-light amplitude receives a positive addition $\propto R^3$; 6.5×10^{-5} of QED for a proton-scale core	VII	a measured deficit, or an excess of the wrong sign
exactly two massless transverse director modes	VII	a scalar or longitudinal polarisation in the same sector
odd Q_H quantises as a fermion, even Q_H as a boson	VI	an even-charge fermionic soliton
$E_Q < Q\,E_1$: multi-charge defects do not fission	IV	an observed fission of a $\lvert Q\rvert > 1$ state
no isolated configuration of non-zero triality	VIII	a free quark
charge is quantised by the integer winding of the Hopf fibre	VIII	non-integer charge, or a magnetic monopole in this sector
generations cannot arise in the director sector alone	VIII E	an S^2 director model producing three generations
the soliton's internal oscillation is a resonance of hadronic width, $\Gamma/\omega = 0.12$	VIII E	a narrow vibrational state in a gapless director medium
topological protection requires rigidity at the cutoff	IV C	a protected topological code in a medium with a soft cutoff-scale mode